A Continuous Benchmarking Community Challenge for COVID-19 Outcome Prediction

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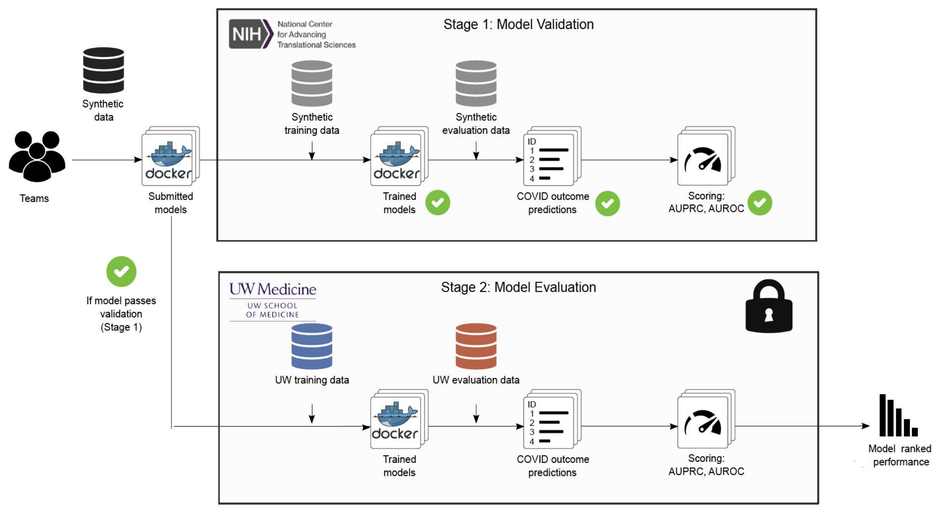
Implementing a Model-to-Data approach to enable citizen science and model benchmarking for COVID-19 outcome prediction

Alternative approach for sensitive EHR data sharing

Sensitive EHR data sharing is restricted under HIPAA to protect patient privacy, raising the threshold for researchers outside health institutions to access and utilize patient data to aid in the COVID-19 pandemic response.

Our solutions - Model-to-data paradigm

- Containerized models submitted by participants are trained and evaluated on COVID-19 patient EHR datasets from UW Medicine health system in secure UW on-premises computing environment.
- Only model performance scores Area Under the Receiver Operator Curve (AUROC) and Area Under Precision-Recall Curve (AUPRC) are returned to participants



EHR DREAM Challenge - COVID-19 Prediction

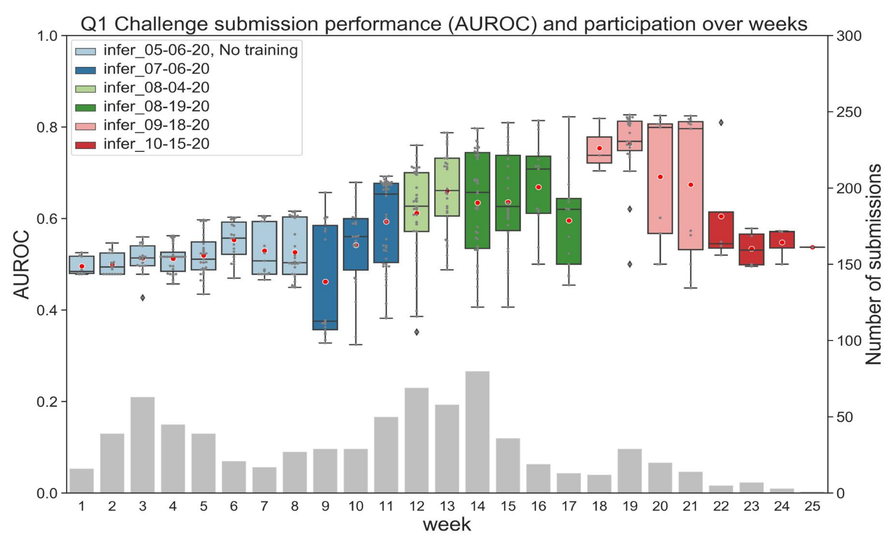
Questions

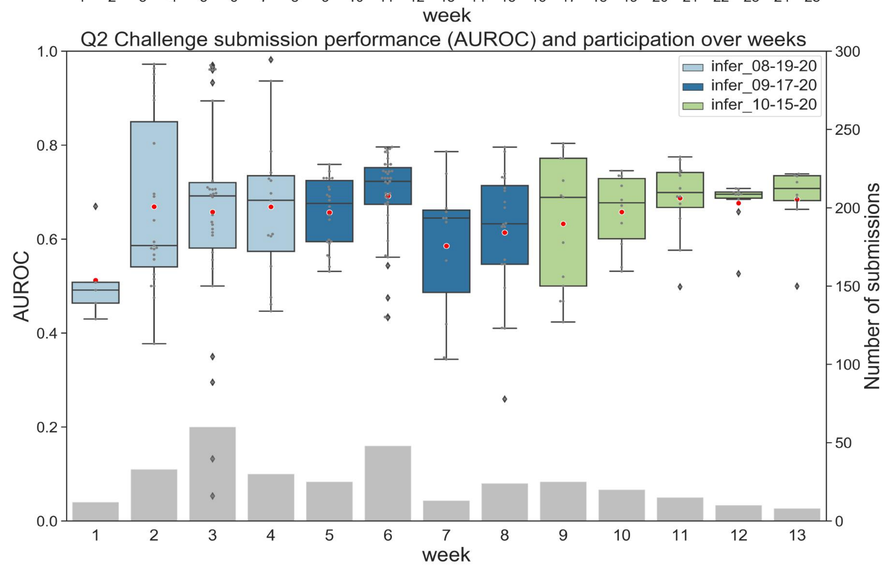
Q1. For a patient who has received a COVID-19 test, how likely will he/she test positive?

Q2. For a patient who has tested positive during an outpatient visit, how likely will he/she be hospitalized within 21 days from COVID-19 test?

Data

- Data are updated every **2-4** weeks. Six releases of the dataset in Q1 have been prepared over 25 weeks. The dataset in Q2 has been updated 3 times over 13 weeks.
- The data were collected from the UW Medicine health system and were dated back to 2010.
- As of late November 2020, the combined dataset included ~ **90,000** patients tested for COVID-19 since February 2020. ~ **3,500** patients have tested positive. **~ 120** have been hospitalized within 21 days after testing positive.





Participation

- As of late November 2020, **482** participants have registered for the challenge.
- We received **483** valid submissions from **65** teams for Q1 and **226** valid submissions from **25** teams for Q2
- The AUROC and AUPRC of the best-performing model so far are **0.827** and **0.303** seen in the dataset version "09-18-20" for Q1, and **0.982** and **0.897** for Q2 dataset version"08-19-2020".

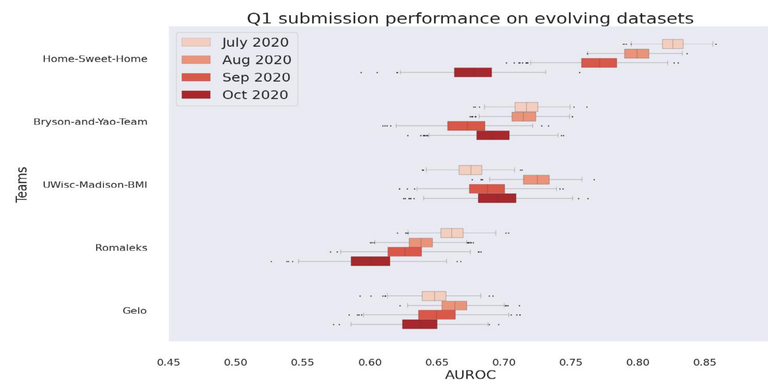
Post challenge analysis

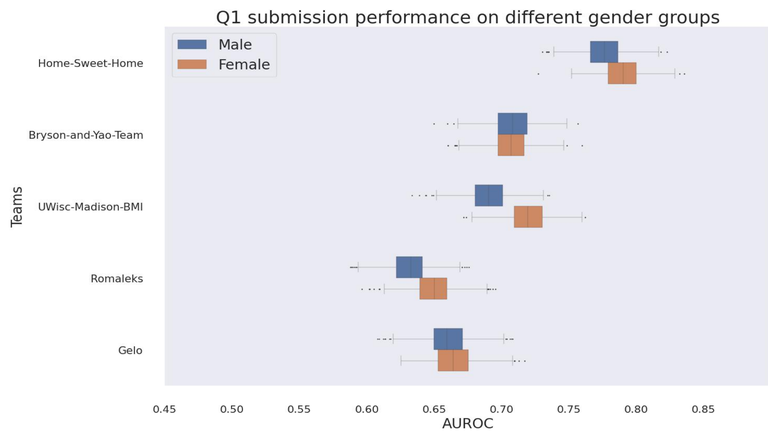
Experiment design

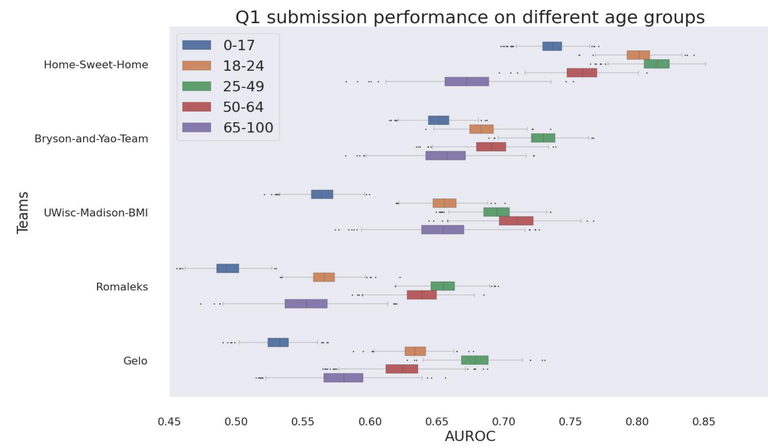
As of mid November, our accumulated datasets included ~ 90K patients who have tested for COVID-19. We split the datasets into training (patients who tested before July 2020, 50%, ~ 45K patients) and evaluation datasets (patients who tested after July 2020, the other 50%). We selected the top 5 models from the challenge Q1 and re-trained the models on the training datasets and evaluated the models using the evaluation datasets. We used bootstrapping to get models' performance on sub-datasets stratified by months when patients were tested, age and gender.

Conclusions

- Some models might overfit to the training data and model's performance degrades over tem -poral evolving datasets while some are more resistant to performance degradation over time (UWisc-Madison-BMI).
- Four out of the five best-performing models achieve a significantly higher AUROC value on female patients compared to male patients ($p < 0.0001$).
- The performance of models is the highest on the 25-49 age group among all other age groups.







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Reference

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